Zackary Blaylock

Senior Lead Software Engineer

PROFESSIONAL SUMMARY

Senior lead software engineer with 8+ years building secure, cloud-native systems on AWS. Designed and built three production systems at SendSafely: the Actions workflow engine, an MCP server that connects AI assistants such as Claude to encrypted files, and a standalone OAuth 2.1 authorization server. Previously built payment and eCredit redemption services at Delta Air Lines. Full-stack across TypeScript, Java, Rust, React, and Angular, with a focus on security, end-to-end encryption, and serverless architecture.

RELEVANT WORK EXPERIENCE

SendSafely, Remote

End-to-end encrypted secure file transfer platform

Senior Lead Software Engineer 12/2020 – Present

- Lead full-stack development across web applications, client SDKs, and third-party integrations built on end-to-end encryption (E2EE) principles.
- Designed and built SendSafely Actions as the sole engineer: a serverless workflow engine on AWS Lambda that moves event processing off the core web application and lets customers plug in their own processing, such as DLP and antivirus scanning, through signed webhooks. Supports both org-wide and personal workflows executed in priority order, running in production across US, EU, and AU regions.
- Initiated and built SendSafely's MCP integration, now live for customers, which lets AI assistants such as Claude list, inspect, and manage encrypted packages and workspaces. Built both a local stdio server that handles encryption on the user's machine, with approval prompts before destructive actions, and a stateless remote connector on AWS Lambda that only sees metadata, so file contents and keys never leave the client.
- Architected and built SendSafely's OAuth 2.1 authorization server as a standalone microservice (Koa on Lambda, Aurora MySQL, Angular consent UI) with PKCE, token exchange, rotating refresh tokens with reuse detection, and KMS-signed JWTs; integrated bearer-token validation and per-endpoint scope enforcement into the platform REST API and configured the client applications that connect to it.
- Designed and deployed Amazon Bedrock agents that automate form building and cut manual configuration.
- Provision and maintain AWS infrastructure with CloudFormation across ECS, Lambda, EC2, RDS, DynamoDB, S3, and CloudFront, secured with IAM, WAF, and Secrets Manager and monitored with CloudWatch and SNS.
- Built and shipped integrations for Freshdesk, Zendesk, Salesforce, Outlook, Slack, Chrome, and a serverless email gateway, and maintain client API SDKs in Java, Node.js, Python, and .NET.
- Created Claude agent skills, adopted across the engineering team, that standardize development workflows.
- Enhanced CI/CD pipelines with GitHub Actions, CodeArtifact, and AWS deployment strategies, improving release consistency and reducing manual deployment steps.
- Partner with Product Management on technical planning and

CONTACT

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SKILLS

Languages:
TypeScript, JavaScript, Java, Python, Rust, SQL, Bash

Frontend:
React, Next.js, Angular, Vue, Redux, Tailwind CSS, SASS, Astro, WCAG accessibility

Mobile & Desktop:
React Native (Expo), Tauri

Backend:
Node.js, Spring Boot, Spring Security, Spring Data JPA, Hibernate, Koa, Axum (Rust), Prisma

Databases:
PostgreSQL, MySQL, Aurora MySQL, MongoDB, DynamoDB, DocumentDB, Drizzle ORM, Flyway

AWS:
ECS, App Runner, Lambda, EC2, VPC, ALB, RDS, S3, CloudFront, Route 53, Bedrock, IAM, WAF, Secrets Manager, KMS, API Gateway, SSM, SES, SNS, EventBridge, DynamoDB Streams, CloudWatch, ECR, SAM, CodeBuild, Amplify

Other Cloud:
Google Cloud (GCP), Firebase, Heroku

DevOps & CI/CD:
Docker, Docker Compose, CloudFormation, GitHub Actions, Jenkins, CodeArtifact

Auth & Security:
OAuth 2.1, PKCE, Token Exchange (RFC 8693), JWT, Auth0, Auth.js, HMAC request signing, E2EE

Payments & Messaging:
Stripe, PayPal, Alipay, Apple

roadmap execution, and lead triage of customer-reported defects through fix and release.

Delta Air Lines, Atlanta, GA
Full Stack Software Engineer 03/2019 – 12/2020

- Developed and maintained a payment processing engine built on Spring Boot REST APIs and microservices, supporting all major card networks (American Express, Visa, Mastercard, Discover, Diners Club, JCB, UnionPay, and UATP) plus PayPal, PayPal Credit, Alipay, Apple Pay, and instant bank transfer, including CID/CVV and UnionPay SMS passcode verification and PSD2 compliance.
- Built the backend for redeeming Delta eCredits, eCertificates, Gift Cards, and other e-documents as payment on delta.com, partnering closely with the frontend and eligibility-rules teams to deliver the feature quickly and reliably (Certificates, eCredits & Gift Cards).
- Led development of an Angular payment portal UI integrating multiple payment methods, ensuring WCAG accessibility compliance.
- Maintained Jenkins and Gradle CI/CD pipelines with JUnit and Jest test coverage, partnering with QA on test automation.
- Provided production support and break-fix response for critical payment defects affecting customer transactions.

PREVIOUS EXPERIENCE

Frontend Software Developer, Cox Automotive, Remote 01/2019 – 03/2019
Software Developer, Hiliteu (Startup), Remote 01/2018 – 01/2019

PROJECTS

SendSafely Actions: Multi-tenant workflow automation for an E2EE file-sharing platform. TypeScript Lambdas build an execution plan for each event, run native actions through the platform REST API, and pause on HMAC-signed webhooks to customer services (antivirus, DLP, archiving) until they call back, with 15-minute timeouts, replay of failed runs, and SES notifications. Configuration is stored in MySQL (Hibernate), and execution state in DynamoDB, streamed to DocumentDB for customer-facing logs. Deployed with AWS SAM and CodeBuild. Also built the proof-of-concept OPSWAT antivirus webhook receiver.

SendSafely MCP Server: TypeScript MCP server (npm workspaces monorepo) shipped two ways: a published local stdio package with OS-keychain credentials and client-side encryption, and a stateless Streamable HTTP connector on Lambda behind CloudFront, WAF, and API Gateway that exchanges the user's OAuth token for a tenant-scoped token on every call. Packages are published through GitHub Actions with OIDC to CodeArtifact.

SendSafely OAuth Authorization Server: OAuth 2.1 microservice (Koa on Lambda behind CloudFront, Aurora MySQL, Angular sign-in and consent app) supporting PKCE, resource indicators, token exchange, introspection, and revocation. Users sign in through the main application via a KMS-signed handover, tokens and client secrets are hashed at rest, and Java servlet filters validate bearer JWTs against JWKS and enforce scopes on every API endpoint.

Delta eCredits, Certificates & Gift Cards: Backend services (Java, Spring Boot) that let customers apply eCredits, eCertificates, Gift Cards, and other e-documents as payment when booking on delta.com, built alongside the frontend and eligibility-rules teams (Certificates, eCredits & Gift Cards).

Delta Payment Engine: Backend payment services (Java, Spring Boot REST APIs and microservices) for delta.com checkout, supporting all major card networks (American Express, Visa, Mastercard, Discover, Diners Club, JCB, UnionPay, UATP) plus PayPal, PayPal Credit, Alipay, Apple Pay, and instant

Pay, UnionPay, UATP, Fiserv CardPointe, card networks (Visa, Mastercard, Amex, Discover, JCB, Diners Club), PSD2, Twilio

Testing:
Jest, Vitest, JUnit, Jasmine, Karma

AI:
Amazon Bedrock Agents, Anthropic API, OpenAI API, OpenRouter, Claude Code, Cursor, MCP

Architecture:
REST APIs, OpenAPI, Microservices, Multi-tenancy

Tools:
Git, GitHub, GitLab, npm, Yarn, Maven, Gradle, Cargo, Vite, Webpack

EDUCATION

University of Kentucky
Bachelor of Business Administration
Management
Lexington, KY
05/2016

Kennesaw State University
Certificate in Computer Science Foundation
Kennesaw, GA
05/2017

Thinkful
Full Stack Web Development Immersion (Bootcamp)
03/2018

bank transfer, with CID/CVV and UnionPay SMS passcode verification and PSD2 compliance. Maintained and hardened the engine in production, including break-fix response for critical payment defects.

PQVault *(personal side project, in development):* Post-quantum encrypted password manager with desktop (Rust, Tauri), browser extension, web, and mobile (React Native, Expo) clients built on a shared React/TypeScript UI library. Rust (Axum) API on PostgreSQL with AWS infrastructure defined in CloudFormation. Uses ML-KEM-768 for device keys and native biometric unlock (Touch ID, Windows Hello).